

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 7 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTCVE705A - ENGINEERING ECONOMICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
-

Q1. Objective type questions. (Compulsory Question) 12

- 1 The Study of Engineering Economics mainly deals with a) Production onlyb) Demand and supply, cost, value of moneyc) Machine designd) Thermodynamics
- 2 When demand is perfectly elastic, the elasticity value is:a) Zero b) One c) Infinity d) Negative
- 3 What is the most important concept in engineering economy a) Demand and supply b) Types of cost c) Time value of money d) Balance sheets
- 4 Continuous compounding interest formula is: a) $A = P(1+i)^n$ b) $A = P e^{rt}$ c) $A = P/\eta d$ d) $A = P - r$
- 5 Parametric estimation is based on: a) Random trials b) Statistical correlation of parameters c) Market demand d) Annual budgeting
- 6 Depreciation is caused due to: a) Wear & tear b) Obsolescence c) Accidents d) All of the above
- 7 A systematic allocation of the cost of a tangible asset over its useful life is called a) Taxation b) Depreciation c) Investment evaluation d) Working capital
- 8 Which method of depreciation gives maximum depreciation in the first year? a) Straight line b) Double declining c) Sinking fund d) Unit of production
- 9 What does the gradient factor $(A/G, i, n)$ convert? a) A gradient series into present worth b) A gradient series into future worth c) A gradient series into uniform annual series d) A uniform series into gradient series
- 10 The financial ratio useful for liquidity is: a) Debt ratio b) Current ratio c) Return ratio d) Activity ratio
- 11 Capital recovery factor (A/P) is used to find: a) Annual equivalent of present value b) Future value of annuity c) Present value of gradient d) Depreciation
- 12 The depreciation balance method is also known as a) Straight line method b) Written down value method c) Sum of the year's digit d) Unit of production method

Q2. Solve the following.

- A) Explain demand and supply curves with neat diagram.**

6

- B) Discuss the classification of cost in detail and develop a cost - time relation ship with an example. 6
- Q3. Solve the following.
- A) Drive the formula for uniform series present worth(P/A).also give engineering applications. 6
- B) Differentiate between nominal interest rate and effective interest rate. derive the formula for effective interest rate. 6
- Q4. Solve Any Two of the following.
- A) Explain management accounting. how does it support decision making in engineering projects. 6
- B) Explain lease financing with advantages, limitation and evaluation criteria. 6
- C) What are financial accounting principles? explain any four basic accounting principles with examples. 6
- Q5. Solve Any Two of the following.
- A) Explain the concept of cost of capital. how it is used in making financing decision for projects? 6
- B) What is taxation in engineering economics? explain different types of taxes applicable to engineering projects 6
- C) Explain in detail the concept of risks in project financing and how they are managed. 6
- Q6. Solve Any Two of the following.
- A) Explain any two depreciation methods in detail. compare their effect on book value and tax liability. 6
- B) What is profit and loss statement. explain its structure and importance in evaluating profitability. 6
- C) What is international finance? explain factors influencing international financial flow. 6

***** End *****